

YADHU NANDHAN T.J.

AI & Full-Stack Developer

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SUMMARY

MCA student and developer who builds AI-integrated full-stack applications end to end — RAG pipelines over vector databases, real-time computer vision, and Flask/FastAPI backends — owning each project from schema design through API layer to deployed frontend.

TECHNICAL SKILLS

Languages	Python, JavaScript, TypeScript, SQL, HTML/CSS
Backend	Flask, FastAPI, REST APIs, JWT authentication, Socket.IO, Pydantic, Uvicorn
AI & CV	Google Gemini API, LangChain, RAG pipelines, Pinecone, ChromaDB, MediaPipe, OpenCV
Frontend	React, Next.js, Three.js, Tailwind CSS, Vite
Databases	PostgreSQL, MySQL, MongoDB (Beanie ODM), SQLite, schema design
Tools	Git, GitHub, Pytest, Jinja2, bcrypt
Spoken	English, Malayalam

PROJECTS

NestQuest — AI Rental & Real Estate Matchmaker | *Flask, PostgreSQL, ChromaDB, LangChain, Gemini*

- Designed a dual-database architecture separating transactional PostgreSQL records from a ChromaDB vector store, making unstructured broker notes a first-class searchable data source.
- Built a RAG pipeline that answers plain-language queries such as “a quiet apartment near Infopark for night-shift work” and generates grounded explanations of why each property matches.
- Implemented JWT authentication and bcrypt-backed role-based access for Admin, Broker and User roles, with broker listing dashboards and visit-booking workflows.

Motion Link — Full-Body Gesture-Controlled 3D Game | *MediaPipe, OpenCV, Three.js, Socket.IO, Gemini*

- Built a real-time pose pipeline detecting walk-in-place, body lean and arm raises, auto-calibrating to each player’s baseline motion and streaming telemetry over Socket.IO to a procedurally generated Three.js maze — no keyboard or controller anywhere in the loop.
- Integrated Gemini for dynamic quiz generation with an offline fallback; demonstrated live at an exhibition and drew acquisition interest from an AI/robotics company.

S-TORE — AI-Powered E-commerce Platform | *Flask, Next.js, Pinecone, LangChain, Gemini*

- Implemented vector product discovery with Pinecone and sentence-transformer embeddings, letting shoppers search by description or use case rather than exact keywords.
- Built a conversational shopping assistant with tool orchestration over cart, order and recommendation APIs, backed by a JWT-secured Flask REST layer.

Automated-MCA — Department Portal & Auto-Greeter | *FastAPI, MongoDB, React, TypeScript*

- Built a FastAPI backend with Beanie ODM over MongoDB, Pydantic validation, token-based admin auth and a Pytest suite over the API routes.
- Added Gemini-generated greeting templates behind a pending/approved moderation workflow, with certificate generation and CSV member import/export.

Also built: Thought Tracker (React 19 + TypeScript, GitHub Gist as an offline-first datastore), a GSAP/Lenis portfolio site, and a React/Vite storefront.

EXPERIENCE

Web Developer — Freelance | *Two client engagements* 2023 – 2025

- Delivered a business website and an art e-commerce store for two local clients, working directly with their teams through iterative review and troubleshooting cycles.
- Collaborate with UC College, Aluva on internal software projects for the department, taking them from requirements through to deployment.

EDUCATION

Master of Computer Applications (MCA), UC College, Aluva	2025 – 2027
Bachelor of Computer Applications (BCA), NSS College, Rajakumari	2022 – 2025
Higher Secondary Education (Class XII), HSS Poothotta	2020 – 2022
SSLC (Class X), KPMVHSS Poothotta	2019 – 2020